

Yashasvi Kumar

Ranchi, Jharkhand , India

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EDUCATION

National Institute of Technology , Agartala

B.Tech – Production Engineering

Aug 2023 – Present

CGPA: 7.93/10.00

Jawahar Vidya Mandir,Shyamali,Ranchi,CBSE

Mar 2022

Senior Secondary Education (Mathematics/Physics/Chemistry)

Percentage: 80.6%

St.Michael's School,Ranchi,CBSE

July 2020

Higher Secondary Education/Grade 10

Percentage: 96.4%

RESEARCH EXPERIENCE

Research Intern — Shock Waves and High-Speed Flow Lab, IIT Jodhpur [Certificate] May–Jul 2025

- Performed CFD simulations of cavity flow for efficient fuel–air mixing in SCRAMJET engines using ANSYS Fluent.
- Designed and implemented a sinusoidal actuation mechanism using a closed-loop hybrid servo motor for aerodynamic pressure variation studies.
- Conducted analytical study for burner design for supercritical kerosene injection in SCRAMJET engine.

PROJECTS

1. Hybrid Battery Thermal Management Simulation using PCM + Liquid Cooling (Personal Project) -Sep 2025

- Designed and simulated a hybrid battery thermal management system (BTMS) integrating phase change material (PCM) and liquid cooling channels in ANSYS Fluent.
- Analyzed transient temperature profiles and optimized coolant flow parameters for improved thermal uniformity.
- Compared temperature regulation efficiency of hybrid configuration with conventional liquid cooling, demonstrating enhanced heat absorption capacity and reduced peak cell temperature.

2.Review Paper — Hybrid Battery Thermal Management using PCM + Liquid Cooling-(Ongoing, NIT Agartala)

- Conducting a comprehensive literature review on hybrid BTMS configurations combining PCM and liquid cooling for EV battery packs.
- Analyzing comparative studies on PCM selection, channel design, and coolant flow patterns for efficient temperature control and cell balancing.
- Preparing a detailed review manuscript summarizing advancements, challenges, and future research trends in hybrid cooling of lithium-ion batteries.

TECHNICAL SKILLS AND CERTIFICATIONS

Software-Tools: Ansys-Fluent, Comsol Multiphysics, Matlab, SolidWorks, Icem-CFD

languages:C++,Python

MOOCs:

- Heat Exchangers: Fundamentals and Design Analysis-NPTEL-ONGOING
- Convective Heat Transfer-NPTEL-ONGOING

ACHIEVEMENTS

- Ranked among the top 15 students in the Department of Production at NIT Agartala based on cumulative academic performance.
- Received Certificate of Merit from CBSE for outstanding academic excellence and being among the top 0.1% of successful students in Science (Theory) nationwide in Class 10.
- Received official appreciation from IIT Jodhpur for exemplary research performance during internship at the Shock Waves and High-Speed Flow Lab (SWAHS Lab) under Dr. Arun Kumar R.[Certificate]